

Shayan Poigai

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EDUCATION

Santa Clara University — B.S. in Computer Science

Expected Fall 2027

Major GPA: 4.0

WORK EXPERIENCE

Co-Founder & CTO/Lead Software Engineer

May 2026 – Present

Adorus (adorusjewels.com)

Scope & ownership

- Sole developer on a 4-person founding team, building a full-stack jewelry e-commerce platform end-to-end across 6 planned development phases (pre-launch).
- Translated co-founder feedback into data models, REST API contracts, and component architecture spanning 33 Flyway migrations, 35+ API endpoints, and a ~32-product catalog with color-variant support.

Build-vs-buy decision

- Chose a custom AWS build over Shopify, self-hosting the entire platform for ~\$40/month vs. an equivalent Shopify plan + paid apps for try-on, reviews, analytics, and returns (~\$250+/month) — an ~80% (~\$2,500/year) cut in recurring platform costs, while avoiding Shopify's per-transaction markup by settling directly through Stripe.

Architecture & infrastructure

- Built a responsive React (Vite) frontend on Java Spring Boot REST services and PostgreSQL (AWS RDS), deployed across 5 AWS services (EC2, RDS, S3, CloudFront, Cognito) with GitHub Actions CI/CD and health-checked backend releases.
- Architected a private product-image pipeline: S3 bucket with all public access blocked, served through CloudFront with Origin Access Control, plus cache invalidation on product deletion so removed assets stop serving immediately.
- Provisioned least-privilege IAM access for the backend, scoping a custom policy to object-level operations on a single bucket rather than broad managed policies.
- Implemented AWS Cognito auth with 2 role tiers (customer, admin).

AI – virtual try-on

- Designed and shipped an AI virtual try-on feature (beta) rendering a photorealistic preview of a customer wearing the exact product from an uploaded photo or live webcam capture in ~8 seconds via Google Gemini — a differentiator most small jewelry retailers lack.
- Benchmarked 5+ image-generation approaches (AWS Bedrock Nova Canvas & Stability models, AWS Rekognition landmark compositing, Google Gemini) on identity preservation, product fidelity, latency, and cost (~\$0.05/image) before selecting a reference-conditioned model.
- Engineered measurement-driven prompting that parses per-product dimensions (chain length, pendant size) into true-to-life scale and physically accurate necklace drape, with per-product overrides.
- Shipped the try-on stack end-to-end: auth-gated Spring Boot endpoints, a 3-generation free-credit quota that resets on purchase, and a React widget with photo upload, live camera capture (`getUserMedia`), image download, and feedback — storing zero customer photos.

Payments, fulfillment & inventory

- Integrated Stripe for PCI-compliant checkout, automated sales-tax collection, and webhook-driven fulfillment.
- Automated the full post-purchase lifecycle with server-verified Shippo tracking webhooks that auto-advance order status across 6 states, triggering branded transactional emails via Resend.
- Implemented prepaid return shipping labels via the Shippo API with emailed carrier labels and real-time tracking in order history.
- Built per-product and per-variant inventory tracking with admin editing and automatic stock decrement on paid orders.

Analytics & compliance

- Integrated Hotjar product analytics behind a GDPR/CCPA-compliant, opt-in cookie-consent system (3-tier consent + PII masking).

Debugging & judgment

- Diagnosed and fixed a tax-accounting bug (tax folded into shipping) by sourcing authoritative totals from Stripe, backed with 3 regression tests; proactively hardened a migration against a cross-environment production failure.
- Authored a blog post on human oversight of AI coding tools and safe data-migration practices, prompted by a production migration incident.

Software Engineering Intern

June 2026 – Present

HerbsPro

- Implementing backend features across the e-commerce stack.

Machine Learning Research Assistant

Dec 2025 – June 2026

Santa Clara University

- Co-authored a peer-reviewed paper as primary writer, accepted at HIBIBI 2026 (ASONAM-collocated), publishing with Springer.
- Built an NLP/data pipeline analyzing 5,000+ scraped posts from Twitter, Reddit, and YouTube to study public sentiment toward LLMs as a therapeutic tool, applying BERTopic, LDA, and NMF topic modeling.

- Engineered Pandas preprocessing to normalize unstructured text, reducing noise in high-dimensional social media datasets.
- Produced Matplotlib / Seaborn analytics on shifting attitudes toward AI mental-health support.

CS Teaching Assistant & Algorithms Grader

Sept 2025 – June 2026

Santa Clara University

- Mentored 40+ students in C++ (memory management, pointers, OOP) and debugged logic errors live using GDB
- Graded Theory of Algorithms coursework for 40+ students across 2 sections, evaluating runtime analysis, greedy, divide-and-conquer, and dynamic-programming solutions.

PROJECTS

Disease Tracker (AWS Hackathon)

Oct 2025

React, FastAPI, AWS Bedrock, DynamoDB

[Github](#)

- Built a FastAPI REST API serving real-time disease metrics and risk assessments with low-latency responses.
- Engineered an AWS Bedrock AI risk-scoring pipeline and designed DynamoDB schemas for scalable data ingestion.

Kuhn Poker vs Quantum

June 2026

Qiskit, IBM Quantum, FastAPI, React, Vercel, Render

[Github](#), [Live](#)

- Built a full-stack game where a 12 parameter, 6-qubit variational quantum circuit (superposition + entanglement) derives the opponent's mixed strategy, deployed on Vercel and Render.
- Ran the circuit on real IBM quantum hardware (e.g., `ibm_fez`, `ibm_mukesh`); on a 4,096-shot run, real-chip probabilities matched simulation within ~1.7 percentage points on average (max ~4), the gap attributable to quantum noise.

Agentic Restaurant RAG

July 2025

Python, ChromaDB, Ollama, MCP, RAG

[Github](#)

- Built a RAG system answering natural-language restaurant questions via ChromaDB semantic search.
- Exposed data as agent tools over MCP; tested on OpenWebUI with Ollama.

TECHNICAL SKILLS

Languages: C++, Python, Java

Frameworks & Tools: React.js, Node.js, Spring Boot, FastAPI, Docker, Linux, Qt, Git, Vercel, CI/CD, Render

Cloud & Data: AWS (EC2, Bedrock, Cognito, S3, Cloudfront), PostgreSQL, DynamoDB, ChromaDB, Hotjar

AI / ML: RAG, MCP, NLP, Ollama, OpenWebUI, NumPy, Pandas, Matplotlib, topic modeling (BERTopic, LDA, NMF)

Others: Qiskit, IBM Quantum, Claude Code